

Introduction to relational databases

Tables

Relational databases organize data into tables

Tables can be linked together

A relation is a table

When a table is a relation?

ISBN	Title
7654321123456	Creating relational databases for fun and profit
9876543212345	Relational databases for really, really smart people
3212345678909	My life with relational databases: a memoir
8172635412345	Relational databases: an existential journey

Figure 1: Some sample books

Table organization

Each row in the table describes a single book

The data is organized into columns

Each *entry* (or cell) contains **a single piece of data**.

How do we handle a book with two authors?

Table 1: books with multiple authors

ISBN	title	Authors
62112346	Creating relational databases for fun and profit	Lopez Baranda Christina, Jones Hannah, Turay Tandice
84321235	Relational databases for really, really smart people	Novak Stanislaw, Turay Tandice
64567899	My life with relational databases: a memoir	Roy Shanta
87261235	Relational databases: an existential journey	Khatami Paree

Not a good idea

Linked tables

- One cell, one data
- One table, one set of similar facts/situations

Two tables can be linked to obtain more information.

Needs an identifier (ID) for the each row (**primary key**)

Table 2: books

book_id	ISBN	title
1	7654321123456	Creating relational databases for fun and profit
2	9876543212345	Relational databases for really, really smart people
3	3212345678909	My life with relational databases: a memoir
4	8172635412345	Relational databases: an existential journey

Table 3: Authors

author_id	last_name	first_name
1	Lopez Baranda	Christina
2	Jin-Soon	Sin
3	Jones	Hannah
4	Novak	Stanislaw
5	Turay	Tandice
6	Roy	Shanta
7	Berger	Henry
8	Khatami	Paree

Table 4: BooksAuthors

book_id	author_id
3	6
2	4
2	5
1	1
1	3
1	5
4	8

Bridging table

Tables, relationships, and IDs

- intermediate table: *relation* or *join* or *bridging* table
- each row must be referenced uniquely
- so that we can reference a row *from* a different table
- join tables are not necessary to refer to another table — e.g. one-to-many relations

Table 5: Editions

edition_id	book_id	date_of_publication	edition_number
1	3	2001	1
2	3	2003	2
3	4	2003	1
5	1	2000	1
6	3	2005	3
8	2	2012	1
9	3	2009	4

Standard table with reference to another table: **foreign key**

A goal of DBMS is to translate **data** into **information**

- Structured data are more informative
- Constraints are a form of structure
- Examples: instance consistent with schema, foreign key, $\text{day} \leq 31$

NULL values

- Whenever we don't have a value, we can put a NULL
- Can mean:
 - Value does not exists
 - Value exists but is unknown
- Introduce flexible schema
 - First Name, *Second name*, Surname
- Neither TRUE nor FALSE

Data Anomalies

Teacher	Course ID	Course Name
Mary Smith	3	Calculus
Ann Brown	4	Programing
Michael Jordan	4	Programing

- Who teaches a course
- Course name and ID
- What if we fix the typo?
- Can we have a course without teachers?

Functional dependency

Definition: A, B : set of attributes

Then $A \rightarrow B$ if, for any tuples t_1 and t_2 , $t_1[A] = t_2[A] \Rightarrow t_1[B] = t_2[B]$

- $A \rightarrow B$ is a functional dependency
- A functional dependency is information
- A **bad** functional dependency is a problem
 - Goal: remove bad functional dependencies
 - How: change the schema
 - Hurdle: need the instance to have a dependency

Keys and Superkeys

Definition: K : set of attributes of relation R . B is **superkey** of R if for any set B such that $K \cap B = \emptyset$, then $K \rightarrow B$. Equivalent to for any tuples t_1 and t_2 ,
 $t_1[K] = t_2[K] \Rightarrow t_1 = t_2$

Definition: K superkey of relation R . K is **key** if no proper subset of K is a superkey of R

A key is a minimal superkey

Example: Editions table

1. The pair (edition_id, edition_number) of the table Editions is a superkey, since there are no two rows of Editions that have the same values of edition_id and edition_number
2. The pair (edition_id, edition_number) of the table Editions is not a key, since we can remove an attribute (edition_number) and obtain a superkey (the single attribute edition_id)
3. The attribute edition_id is a key. Since it consists of a single attribute, its minimality is trivial
4. The pair (book_id, edition_number) is a superkey. Since removing any of its attribute results in a set of attributes that have duplicates, (book_id, edition_number) is also a key.
5. book_id is a foreign key, since the values of that attribute are found in the key of the Books table.

Boyce-Codd Normal Form

Definition: Let $A \rightarrow B$ for relation R . Then $A \rightarrow B$ is good if A is a superkey of R .

Definition: A relation R is in Boyce-Codd Normal Form if all its functional dependencies are good.

- Boyce-Codd Normal Form is highly desirable
- Boyce-Codd Normal Form not always achievable
- Boyce-Codd Normal Form usually achievable

Normalization

- A bad functional dependency shows which columns must be moved to a new table

Teacher	Course ID	Course Name
Mary Smith	3	Calculus
Ann Brown	4	Programming
Michael Jordan	4	Programming

- Bad functional dependency: $\text{Course ID} \rightarrow \text{Course Name}$
 - New table with attributes Course ID, Course Name
 - Remove Course Name from current table
 - Result: two linked tables

Normalization

Teacher	Course ID
Mary Smith	3
Ann Brown	4
Michael Jordan	4

Course ID	Course Name
3	Calculus
4	Programming

Duplicate rows cannot exist

Planets.csv

```
method,number,orbital_period,mass,distance,year  
Radial Velocity,1,269.3,7.1,77.4,2006  
Radial Velocity,1,874.774,2.21,56.95,2008  
Radial Velocity,1,763.0,2.6,19.84,2011
```

- different separators: space, tab
- rigid structure
- one table per file

JSON file

```
{ "menu": {  
  "id": "file",  
  "value": "File",  
  "popup": {  
    "menuitem": [  
      { "value": "New", "onclick": "CreateNewDoc()" },  
      { "value": "Open", "onclick": "OpenDoc()" },  
      { "value": "Close", "onclick": "CloseDoc()" }  
    ]  
  }  
}}
```

- nested tables
- loose structure

Except where noted, text and images for Introduction to Relational Databases by Mark Jordan is licensed under a Creative Commons Attribution 4.0 International License. Except where noted, text and images for Introduction to Relational Databases by Gianluca Della Vedova is licensed under a Creative Commons Attribution 4.0 International License. Everything in the 'scripts' directory is in the public domain (CC0).